

Report 01 — Project Overview

Project Title: Develop and Test Custom Correlation Rules in a SIEM (ELK Stack) to Detect Credential Stuffing, DNS Tunnelling, and PowerShell Exploitation

Author: Security Engineering Intern **Date:** 2026-07-07 **Version:** 1.0 **Classification:** Internal / Academic Submission

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1. Project Introduction

Modern enterprise networks face a growing variety of sophisticated threats that evade traditional perimeter defences. Credential stuffing attacks exploit stolen password lists to authenticate against public-facing services at scale. DNS tunnelling abuses the universally permitted DNS protocol to exfiltrate data or establish covert command-and-control channels. PowerShell exploitation takes advantage of Windows' built-in scripting engine to execute malicious payloads, often without writing files to disk.

This project implements a Security Information and Event Management (SIEM) solution using the open-source ELK Stack (Elasticsearch, Logstash, Kibana) to ingest, parse, correlate, and alert on log evidence of these three distinct attack categories within a simulated corporate environment named **CORP**.

The project covers the complete lifecycle of SIEM-based threat detection: log source onboarding, data normalisation, custom correlation rule development, alert tuning, and detection validation against realistic log datasets generated on **2026-06-15** across Windows workstations, Windows servers, Linux production servers, network firewalls, and DNS infrastructure.

2. Objectives

#	Objective	Outcome
1	Deploy and configure the ELK	Logstash pipelines parsing

#	Objective	Outcome
	Stack for multi-source log ingestion	Windows Event Logs, syslog, Zeek DNS, and firewall logs
2	Develop custom Elasticsearch correlation rules for credential stuffing	Rule triggers on ≥ 5 authentication failures across multiple accounts from one source IP within 5 minutes
3	Develop custom correlation rules for DNS tunnelling	Rule triggers on high-entropy TXT queries and excessive TXT query volume to suspicious domains
4	Develop custom correlation rules for PowerShell exploitation	Rules trigger on encoded commands, download cradles, and Office-spawned PowerShell processes
5	Validate detection rules against realistic log datasets	True positive alerts confirmed against known-malicious log entries
6	Measure detection accuracy, false positives, and false negatives	Quantified in Report 08
7	Map all detections to MITRE ATT&CK framework	Full mapping in Report 07
8	Produce actionable security recommendations	Documented in Report 09

3. Scope

3.1 In Scope

- All log sources within the `src_data/` directory collected on 2026-06-15
- Windows Security Event Log from domain-joined workstations and servers
- Linux SSH and authentication logs from production servers
- DNS query logs from the corporate DNS server (dns-srv01)
- Zeek network-layer DNS capture
- Windows PowerShell Operational Event Log
- Sysmon process creation logs
- Windows Security process creation (Event ID 4688)
- Network connection telemetry
- Proxy access logs

- Firewall (iptables) allow/deny logs

3.2 Out of Scope

- Physical security controls
- Application-layer vulnerabilities (SQL injection, XSS)
- Email phishing campaigns (only the downstream payload execution is captured)
- Cloud workloads outside the CORP domain
- Encrypted traffic decryption / SSL inspection

3.3 Network Scope

Subnet	Description
10.20.0.0/16	Core infrastructure (DNS: 10.20.0.53)
10.21.0.0/16	Internal workstations and servers
10.22.0.0/16	Additional internal segment
172.16.5.0/24	DMZ / management segment

3.4 Monitored Hosts

Hostname	Type	OS	Role
SRV-DC01	Server	Windows Server	Domain Controller
SRV-APP05	Server	Windows Server	Application Server
SRV-FILE02	Server	Windows Server	File Server
WKS-ENG14	Workstation	Windows 10	Engineering
WKS-FIN07	Workstation	Windows 10	Finance
WKS-HR03	Workstation	Windows 10	HR
WKS-SALES22	Workstation	Windows 10	Sales
bastion01	Server	Linux	Bastion / Jump Host
web-prod01	Server	Linux	Web Server
web-prod02	Server	Linux	Web Server
db-prod01	Server	Linux	Database Server
app-prod03	Server	Linux	Application Server
mail-relay01	Server	Linux	Mail Relay
dns-srv01	Server	Linux	DNS (BIND)
fw-edge01	Appliance	Linux	Edge Firewall (iptables)

4. ELK Stack Overview

The ELK Stack is a widely-adopted open-source platform for centralised log management and security analytics.

Component	Version	Role
Elasticsearch	8.x	Distributed search and analytics engine; stores and indexes all log data
Logstash	8.x	Data collection pipeline; parses, filters, enriches, and forwards logs
Kibana	8.x	Visualisation and alerting UI; dashboards, SIEM app, detection rules
Beats (Winlogbeat, Filebeat, Packetbeat)	8.x	Lightweight agents deployed on endpoints to ship logs
Elastic Agent	8.x	Unified agent managing multiple integrations

4.1 Elasticsearch Index Strategy

Index Pattern	Source
logs-windows.security-*	Windows Security Event Log
logs-windows.powershell-*	PowerShell Operational Log
logs-sysmon-*	Sysmon Event Log
logs-dns-*	DNS server logs + Zeek DNS
logs-auth-*	Linux auth.log / sshd
logs-firewall-*	iptables firewall logs
logs-proxy-*	Squid/proxy access logs
logs-network-*	Network connection telemetry

5. SIEM Workflow

The end-to-end detection workflow follows five stages:

1. **Collection** — Beats agents and Filebeat ship raw logs from all endpoints and infrastructure devices to a central Logstash instance.
2. **Parsing & Enrichment** — Logstash Grok filters parse unstructured syslog and Windows event formats. Fields are normalised to the Elastic Common Schema (ECS). GeoIP and threat intelligence enrichment are applied to external IP addresses.

3. **Indexing** — Parsed, enriched documents are indexed into Elasticsearch with daily rolling indices.
 4. **Correlation & Detection** — Kibana SIEM detection rules (threshold rules, EQL sequences, and custom queries) execute on a scheduled basis against indexed data. Alerts are generated when rule conditions are met.
 5. **Response** — Triggered alerts appear in the Kibana Alerts queue. Analysts triage, investigate, and escalate via a defined incident response workflow.
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6. Architecture Diagram

```
flowchart TD
    subgraph Endpoints["Endpoint Layer"]
        WS[Windows Workstations\nWKS-ENG14, WKS-FIN07\nWKS-HR03, WKS-SALES22]
        SRV[Windows Servers\nSRV-DC01, SRV-APP05\nSRV-FILE02]
        LINUX[Linux Servers\nbastion01, web-prod01/02\nndb-prod01, app-prod03\nmail-relay01]
        DNS_SRV[DNS Server\n dns-srv01]
        FW[Edge Firewall\nfw-edge01]
    end

    subgraph Shippers["Log Shipping Layer"]
        WB[Winlogbeat\nWindows Event Logs\nSysmon / PowerShell]
        FB[Filebeat\nLinux auth.log\nsshd / syslog]
        ZK[Zeek\nDNS / Network Capture]
        FWB[Filebeat\niptables logs]
        PXB[Filebeat\nProxy logs]
    end

    subgraph Pipeline["Logstash Pipeline"]
        LS[Logstash\nGrok Parse → ECS Normalise\nGeoIP Enrich → Output]
    end

    subgraph Storage["Elasticsearch Cluster"]
        ES[(Elasticsearch\nIndexed Documents\nRolling Daily Indices)]
    end

    subgraph SIEM["Kibana SIEM"]
        KD[Detection Rules\nThreshold / EQL / KQL]
        KA[Alerts Queue]
        KV[Dashboards &\nVisualisations]
    end

    subgraph Response["Analyst Response"]
        AN[Security Analyst\nTriage → Investigate\nEscalate / Close]
    end

    WS --> WB
    SRV --> WB
    LINUX --> FB
    DNS_SRV --> ZK
    DNS_SRV --> FB
    FW --> FWB
    LINUX --> PXB
    WB --> LS
    FB --> LS
    ZK --> LS
    FWB --> LS
```

```
PXB --> LS
LS --> ES
ES --> KD
KD --> KA
ES --> KV
KA --> AN
```

7. Environment Summary

7.1 Key User Accounts

Account	Type	Description
jsmith	Standard User	Domain user — Engineering
rgupta	Standard User	Domain user — Finance
kpatel	Standard User	Domain user — Sales
lchen	Standard User	Domain user — Engineering
dwilson	Standard User	Domain user — HR
amehta	Standard User	Domain user — Sales
nreddy	Standard User	Domain user — IT
tverma	Standard User	Domain user — HR
svc_backup	Service Account	Backup service account
svc_sql	Service Account	SQL database service account

7.2 Threat Actor Infrastructure

Indicator	Type	Role
185.220.101.47	IP Address	Tor exit node — SSH brute force / malware C2
103.68.22.9	IP Address	Suspected C2 server
45.146.164.110	IP Address	Scanning / lateral movement attempts
cdn-sync-update.net	Domain	DNS tunnelling C2 domain
malicious-update-cdn.net	Domain	PowerShell payload hosting
http://malicious-update-cdn.net/stage2.ps1	URL	Stage 2 PowerShell payload
http://185.220.101.47/update.exe	URL	Malware binary download
http://cdn-sync-update.net/x.sct	URL	Scriptlet for regsvr32 LOLBAS

7.3 Log Collection Period

Field	Value
Start	2026-06-15 00:00:00 UTC
End	2026-06-15 05:30:45 UTC
Duration	~5.5 hours
Total Log Lines	1,385 across 12 log files